

revolutionising PnP technology. Integrating artificial intelligence (AI) and machine learning (ML) enables machines to learn from patterns, predict maintenance needs, and optimise production processes. This shift reduces downtime and enhances overall equipment efficiency (OEE).

But what does the future hold for AI and ML in manufacturing? The move towards ‘black room technology’—fully automated, robot-assisted production lines—minimises human error and increases efficiency. The industry is shifting towards more efficient, error-free manufacturing by relying on robotics. The integration of Industry 4.0 standards is creating an interconnected ecosystem where machines communicate and make data-driven decisions. How are these advancements shaping the future of manufacturing?

Critical advancements

Enhanced precision and speed. Modern PnP machines have sophisticated servo motors and control systems, allowing for the precise placement of tiny components at unprecedented speeds. Companies like Yamaha and Juki are at the forefront of this technology. Yamaha Motor’s Z:TA-R YSM40R and Juki Corp’s RX-8 are both known for their high placement



Panasonic AM100 modular placement machine
(Credit: <https://djkeurope.com/en>)

rates and compact designs. Neoden’s machine has repeat position accuracy of $\pm 0.01\text{mm}$. ASMPT’s PNP machine has a placement accuracy of $\pm 41\mu\text{m}/3\sigma$. The Yamaha YSM20R is a model that exemplifies this advancement. It boasts a high-speed mounting capability with a maximum speed of 90,000 CPH (components per hour), making it one of the fastest in its class. This is achieved through innovations such as high-speed rotary heads and optimised motion algorithms.

Miniaturisation of PNP heads.

Efforts are being made to create smaller and more efficient PNP heads. This includes experimentation with different types of pumps and suction mechanisms to optimise the pick-and-place process. For example, using micro blowers and adjustments in the pick-up tool area can enhance the machine’s ability to handle different component sizes and weights. The Panasonic AM100 modular placement machine stands out in the realm of electronic component

placement technology, featuring a cutting-edge super multi-14-nozzle head. This innovative design not only emphasises productivity but also versatility, allowing for the efficient mounting of components up to 14mm in size at remarkable speeds.

Improved vision systems. Integrating advanced vision systems has significantly improved component recognition and placement accuracy. With real-time adjustments based on vision system feedback, machines can now handle various

components. The Neoden 8 PnP machine features innovative camera systems for improved calibration and accuracy, while the Mycronic MY300 is noted for its intelligent feeder solution that enhances efficiency and productivity.

Increased flexibility. The modular designs and interchangeable tool heads of modern PnP machines offer unparalleled flexibility, allowing for quick reconfiguration for different tasks. This adaptability is crucial for responding to changing market demands. The ASM SIPLACE TX series, for example, is designed for extreme precision, catering to the demands of industries like electronics where component sizes continue to shrink. These machines use multiple cameras and real-time feedback systems to adjust placements in real time, ensuring high accuracy.

Integration of artificial intelligence. AI transforms PnP systems by enabling them to learn and improve over time. This self-improvement capability marks a significant leap forward in automation technology. The Juki RS-1R incorporates AI to automatically recognise components and adjust its pick and place parameters in real time, which greatly enhances the machine’s setup time



JukiFast smart modular mouter (Credit: <https://www.juki.co.jp>)